HW-05 — Input Modeling: Distribution Fitting and Validation

Module M07 | Chapter 9 | Weight 10% | Difficulty ⋆⋆

Module: M07 Chapter: 9 Weight: 10% Due: End of Week 11

Overview

Fit probability distributions to two empirical clinic datasets and document the choices in a professional input model report. You will use maximum likelihood estimation, goodness-of-fit tests, and a sensitivity check via simulation.

1 Datasets

File	Contents	n
`data/clinic_arrivals.csv`	Inter-arrival times (minutes)	500
`data/clinic_service_times.csv`	Service times (minutes)	500

2 Parts

2.1 Part A — Exploratory Data Analysis (25 pts)

For each dataset compute and report:

- Summary statistics: n, min, Q1, median, Q3, max, mean, std, CV, skewness
- Plots: histogram (30 bins), ECDF, box plot — all in a single 3-panel figure per dataset
- Interpretation: does the shape suggest a particular distribution family? State your hypothesis before fitting.

2.2 Part B — MLE Distribution Fitting (30 pts)

For each dataset fit at least **3 candidate distributions** using MLE (`scipy.stats` or `fitter`). For each fit:

- State the estimated parameters with 95% confidence intervals
- Overlay the fitted PDF and CDF on the empirical histogram and ECDF

- Report AIC and BIC

Candidate families to consider: Exponential, Gamma, Weibull, Lognormal, Normal.

2.3 Part C — Goodness-of-Fit Tests (25 pts)

For your top 2 candidates per dataset:

1. Kolmogorov-Smirnov test: report statistic D, p-value, decision at $\alpha = 0.05$
2. Chi-squared test: report statistic, degrees of freedom, p-value, decision
3. Quantile-quantile (Q-Q) plot against the fitted distribution

Conclude: which distribution do you select for each dataset, and why?

2.4 Part D — Model Selection Memo + Sensitivity Check (20 pts)

1. Write a 1-page **input model specification** (not a narrative — a technical table listing each random input, chosen distribution, parameters, and the evidence supporting the choice).
2. **Sensitivity check:** Run your M05 SimPy clinic model (or any single-server queue) twice: once using the fitted inter-arrival distribution and once with an Exponential having the same mean. Report the resulting 95% CI for W_q in each case. Do they overlap?

Grading Summary

Component	**Points**
Part A — EDA (stats + 3 plots per dataset)	25
Part B — MLE: $\geq$ 3 candidates, AIC/BIC	30
Part C — KS + χ^2 + Q-Q per dataset	25
Part D — Input spec + sensitivity simulation	20
Total	**100**

Full rubric: `rubrics/homework_rubric.pdf` | Solutions: the instructor package (available to instructors on request).